Sowmiya Velmurugan

Data Engineer

+1 (812) 552-8324 | sowmiyah.connect@gmail.com | USA | LinkedIn | Portfolio | GitHub

SUMMARY

Senior Data Engineer with 6 years of experience is architecting enterprise-grade cloud data platforms on Azure (ADF, Databricks, ADLS Gen2, Synapse Analytics) and AWS (Glue, S3, Redshift, EMR), processing 8+ TB of healthcare and logistics data daily using Medallion Architecture, Delta Lake, and PySpark Structured Streaming. Proven track record of cutting pipeline execution time by 40%, achieving 99.9% data accuracy, and delivering HIPAA-compliant, production-grade lakehouse solutions through cross-functional collaboration across the healthcare, insurance, and logistics industries.

SKILLS

Languages & Query: Python, PySpark, Scala, SQL, Spark SQL, Shell Scripting

Cloud Platforms – Azure: Azure Data Factory (ADF), Azure Databricks, ADLS Gen2, Azure Synapse Analytics, Azure Functions, Azure Event Grid, Azure Key Vault, Azure Monitor

Cloud Platforms – AWS: Amazon S3, AWS Glue, AWS EMR, Amazon Redshift, Amazon Athena, AWS Lambda, AWS Step Functions, AWS EventBridge

DevOps & Infrastructure: Terraform, CI/CD, Docker, Kubernetes, Git, GitHub Actions

Data Engineering: ETL/ELT Pipeline Development, Batch Processing, Real-Time Data Processing, Medallion Architecture (Bronze/Silver/Gold), Delta Lake, Delta Live Tables (DLT), Auto Loader, CDC Pipelines, Change Data Capture, PySpark Structured Streaming, Event-Driven Architecture, Apache Kafka, Apache Airflow, dbt, Fivetran, Airbyte, Informatica IICS, Lakehouse Architecture, Data Mesh, DataOps

Data Warehouse & Modeling: Snowflake (Snowpipe, Streams & Tasks, Virtual Warehouses), Amazon Redshift, Azure Synapse Analytics, Star Schema, Snowflake Schema, Dimensional Modeling, Partitioning, Parquet, Delta, ORC, Avro, Apache Iceberg

Spark Optimization: Adaptive Query Execution (AQE), Dynamic Partition Pruning, Broadcast Join Optimization, Predicate Pushdown, Shuffle Tuning, Photon Engine, Query Plan Analysis, Selective Caching

Analytics & SQL: Window Functions, Rollups, Grouping Sets, Analytical Functions, Complex Aggregations, Dimensional Rollups, Performance Metrics, Trend Analysis

Data Quality & Governance: Great Expectations, Monte Carlo, Unity Catalog, Data Lineage, Metadata Management, Schema Validation, SLA Monitoring, Data Contracts, HIPAA Compliance, Audit Trails, MLflow, Feature Store, Feature Engineering

Leadership & Collaboration: Cross-Functional Collaboration, Stakeholder Management, Project Management, Technical Leadership, Mentoring, Agile Scrum, Documentation

PROFESSIONAL EXPERIENCE

Data Engineer, UnitedHealth Group | USA (Remote) — **Oct 2025 – Present**

- Architected production-grade PySpark and Scala pipelines on Azure Databricks ingesting, transforming, and serving 8+ TB of healthcare claims, member, and clinical data daily powering population health dashboards and regulatory reporting for 50M+ members across the UnitedHealth enterprise data platform.
- Designed a cloud-native ingestion framework using Azure Data Factory (ADF), ADLS Gen2, Azure Functions, and Azure Event Grid to consolidate 30+ structured and semi-structured healthcare data sources into a governed Bronze layer, reducing manual integration effort by 60% and accelerating time-to-insight for downstream BI teams.
- Implemented Medallion Architecture (Bronze/Silver/Gold) via Delta Live Tables (DLT) and Auto Loader on Azure Databricks, achieving 99.9% data accuracy, eliminating 3+ hours of daily reconciliation, and enforcing HIPAA-compliant data governance through Unity Catalog with full audit trails and role-based access controls.
- Reduced pipeline execution time by 40% through advanced Spark optimization, Adaptive Query Execution (AQE), Dynamic Partition Pruning, Broadcast Join Optimization, shuffle minimization, and Photon Engine acceleration across high-volume PySpark and Scala workloads.
- Built a data observability framework across 100+ production pipelines using Great Expectations and Monte Carlo, integrated with Azure Monitor and Log Analytics alerting, cutting incident response time from 2+ hours to less than 15 minutes and sustaining data contracts across all reporting layers.

Data Engineer, CGI – Client: Cigna Healthcare | Chennai, India — **Jul 2023 – Sep 2025**

- Delivered scalable PySpark and Scala pipelines on Azure Databricks processing 5+ TB of daily healthcare claims, provider, and member data (HL7 FHIR, EDI 837/835, JSON, Parquet) via ADF and Auto Loader improving analytics data availability by 40% for downstream BI and reporting teams through optimized Delta Lake CDC patterns and schema evolution.
- Built multi-layer ADLS Gen2 Data Lake following Bronze-Silver-Gold Medallion Architecture with Delta Lake schema evolution, Unity Catalog governance, and full data lineage tracking from source to Gold layer across 20+ structured and semi-structured healthcare data sources.
- Implemented Change Data Capture (CDC) pipelines on Delta Lake using merge operations and Kafka-driven PySpark Structured Streaming, reducing data latency from daily batch processing to sub-hourly for key operational datasets and enabling near-real-time analytics for clinical reporting stakeholders.
- Tuned Spark cluster configurations executor memory, shuffle partitions, dynamic allocation, AQE and rewrote inefficient query plans using execution plan analysis, cutting average processing time by 35% and improving cluster resource utilization by 30% at zero additional infrastructure cost.

- Developed dbt transformation models for Silver and Gold layer data marts (Star Schema, Snowflake Schema), standardizing actuarial and compliance business logic and enabling self-service analytics through version-controlled, well-documented dbt pipelines validated by Great Expectations across 10+ Azure Synapse Analytics platforms.

Data Engineer, Cognizant – Client: Hartford Insurance | Chennai, India **Aug 2022 – Jun 2023**

- Built distributed PySpark and Scala ETL pipelines on Azure Databricks processing 3+ TB of daily insurance policy, claims, and actuarial data supporting risk assessment models, underwriting workflows, and regulatory reporting through a governed Lakehouse Architecture.
- Designed dimensional data models (Star Schema, Snowflake Schema) with Snowflake Snowpipe ingestion and Streams & Tasks for near-real-time synchronization, supporting high-performance analytics in Azure Synapse Analytics with partitioning strategies and index optimization.
- Consolidated 10+ enterprise source systems (Oracle, SQL Server, Salesforce) into a centralized ADLS Gen2 data lake via Azure Data Factory pipelines, establishing a single governed data access layer with full lineage that eliminated ad hoc data pulls and cut report generation time by 30%.
- Reduced data quality incidents by 30% by implementing automated row-count validation, null-check routines, referential integrity checks, and pipeline execution logging using Great Expectations, tied to Azure Monitor alerting for same-day incident response.
- Collaborated cross-functionally with actuarial, compliance, and analytics teams across Agile Scrum sprints, driving stakeholder alignment and delivering pipeline features on schedule for regulatory reporting releases.

Data Engineer, IBM – Client: UPS Logistics | Bangalore, India **Jan 2020 – Aug 2022**

- Developed Python, PySpark, and Spark SQL ETL workflows processing 2+ TB of daily logistics and shipment data using AWS Glue, S3, Lambda, EventBridge, and Step Functions implementing event-driven architecture that eliminated 12+ hours of weekly manual processing and fed real-time operational KPI dashboards for 5+ regional teams.
- Designed and maintained Snowflake data warehouse schemas (Star Schema, Snowflake Schema) with Snowpipe ingestion and Streams & Tasks for 15-minute shipment status refresh, integrated with Amazon Redshift via optimized AWS Glue ETL jobs.
- Provisioned reproducible AWS infrastructure using Terraform (IaC), reducing environment setup time from days to less than 2 hours, and championed CI/CD adoption via GitHub Actions for automated data engineering deployments sustaining 99% pipeline uptime across 20+ enterprise workflows.
- Authored comprehensive pipeline architecture documentation, data flow diagrams, transformation rule guides, and operational run books across 20+ workflows using Confluence and GitHub, cutting new engineer on boarding time by 2 weeks and establishing reusable DataOps standards adopted by the broader team.

EDUCATION

Bachelor of Engineering in Computer Science **Aug 2015 – Apr 2019**

K. Ramakrishnan College of Technology, Tamil Nadu, India

PROJECTS

Enterprise Healthcare Claims Lakehouse | *PySpark, Delta Live Tables, Unity Catalog, ADF, Great Expectations, Kafka*
2024–2025

- Architected a Databricks Medallion (Bronze/Silver/Gold) Lakehouse processing 50M+ healthcare claims/month using Auto Loader, ADF, Delta Lake CDC, Spark SQL, and Unity Catalog, delivering HIPAA-compliant pipelines with 99.8% data quality.
- Built Kafka-based PySpark Structured Streaming pipelines with Great Expectations, Azure Monitor, and Monte Carlo for automated validation, SLA monitoring, and sub-hourly CDC, reducing data incident response time by 75%.

DataOps Quality & Observability Platform | *Python, SQL, Great Expectations, Apache Airflow, Docker, GitHub Actions*
2023–2024

- Developed an automated DataOps quality framework using Python, SQL, Great Expectations, and Airflow to enforce schema validation, reconciliation, referential integrity, and data contracts across 15+ production pipelines.
- Containerized and deployed the framework with Docker and GitHub Actions CI/CD, reducing manual QA effort by 60% while standardizing version-controlled data quality across development, staging, and production environments.

CERTIFICATIONS

- Databricks Certified Data Engineer Associate (Databricks)
- Microsoft Azure Data Engineer Associate—LinkedIn Learning (Microsoft)
- AWS Certified Cloud Practitioner—Coursera (Amazon Web Services)